SMART INDIA HACKATHON 2026

TITLE PAGE



- **Problem Statement ID : SIH26032**
- **Problem Statement Title : Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.**
- **Theme : Smart automation**
- **PS Category : Software**
- **Team ID :**
- **Team Name : Trinovex**

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KISAN SATHI : SMART FARMER PROCUREMENT PLATFORM

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01.PROPOSED SOLUTION

(One platform for the complete farmer journey)

- Discover relevant government schemes & procurement opportunities.
- Check eligibility, required documents & procurement details.
- Submit preferred procurement date & time online.
- Track procurement and payment status with timely alerts.

02 . HOW IT ADDRESSES THE PROBLEM

(Making procurement simple and transparent)

- Reduces long queues and unnecessary visits
- Farmer can choose a convenient date & time
- Provides clear step-by-step process guidance
- Enables transparent procurement & payment tracking

03 .INNOVATION & UNIQUENESS

(Farmer-first integrated platform)

- Personalized schemes & service recommendations.
- Simple multilingual farmer-friendly interface.
- Date/time preference + procurement + payment tracking in one workflow.
- SMS/in-app notifications for important updates.

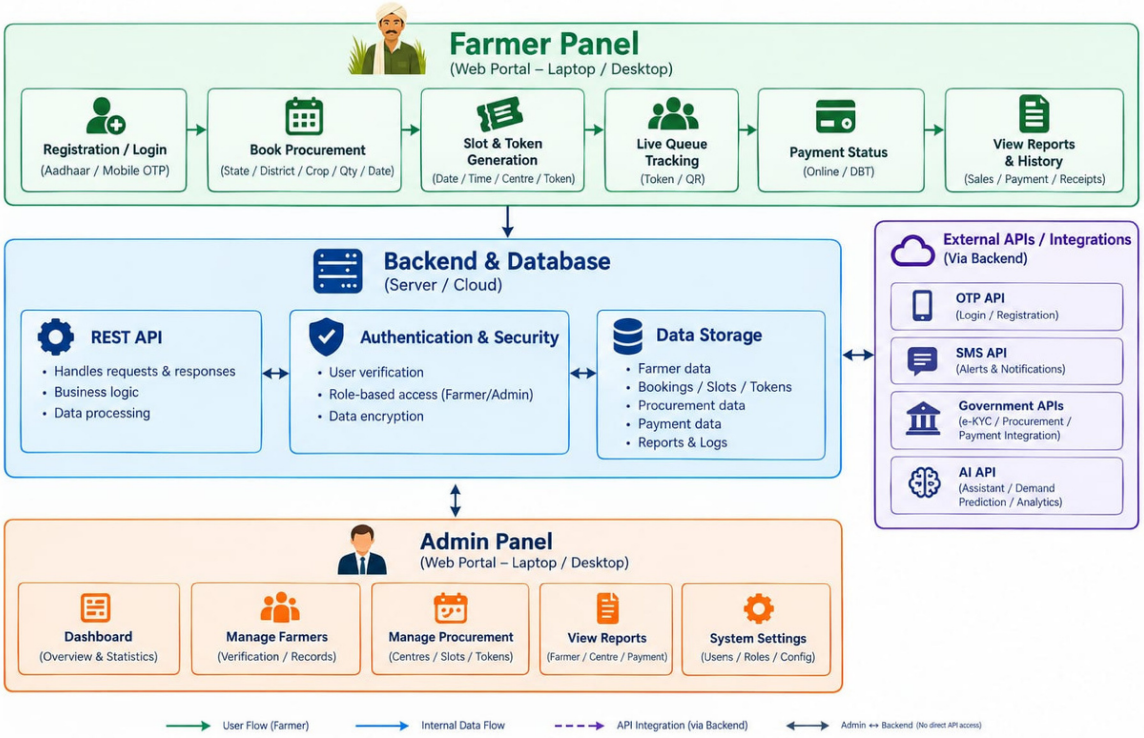
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TECHNICAL APPROACH

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TECHNOLOGY STACK

- **Frontend :** React.js,Tailwind CSS - Responsive Web Interface
- **Backend & Database:** Node.js , Express.js , MongoDB ,Firebase Realtime Database:Booking, Slot & Token Data
- **API & Integrations :** REST API , OTP API(Verification), SMS API(Alarts & Receipt),Ai API(Smart Analytics)
- **Hosting & Security :** AWS,Cloud Deployment ,OTP Authentication HTTPS ,JWT , Role-Based Access



VIEW UI/UX PROTOTYPE

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FEASIBILITY AND VIABILITY

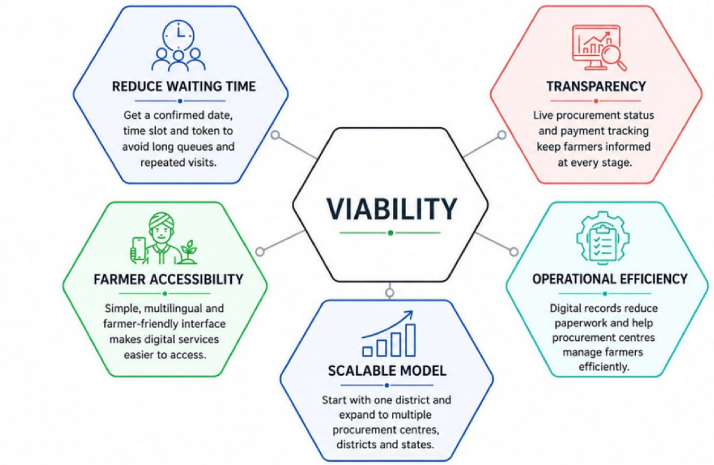
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Cost Feasibility : Uses cost-effective web technologies , cloud infrastructure and open-source tools, making the solution affordable to develop and deploy.

Technical Feasibility : The platform can be implemented using modern web technologies, databases,
secure APIs and scalable cloud infrastructure.

Operational Feasibility : Simple workflows for registration, procurement booking, slot/token management and payment tracking make the system easy for farmers and authorities to use.

Scalability Feasibility : The system can start with selected procurement centres and gradually scale across districts, states and multiple crops.



Potential Challenges

Digital Literacy : Some farmers may face difficulty using digital platforms.

Internet Connectivity : Poor connectivity in rural areas can affect real-time services.

Data Accuracy: Incorrect farmer, crop or procurement data can affect the booking process.

Overcoming Challenges

Simple & Multilingual UI : Provide easy navigation with Hindi/local-language support and minimal steps.

Lightweight + SMS Support : Keep the platform lightweight and use SMS notifications for important updates.

Admin Verification : Use authorized data sources, verification and admin controls to maintain reliable records.

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IMPACT AND BENEFITS

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IMPACT

- **Reduced Waiting Time** — Confirmed date, time slot and token reduce queues and repeated visits.
- **Transparent Procurement** — Farmers can track booking, queue and procurement status.
- **Faster Payments** — Digital payment tracking improves payment visibility.
- **Reduced Manual Work** — Digital records reduce paperwork and repetitive processes.
- **Better Farmer Access** — Schemes and procurement services available through one platform.

BENEFITS

- **Social** — Less waiting and easier access to agricultural services.
- **Economic** — Saves farmers' time and travel costs.
- **Operational** — Helps procurement centres manage slots and queues efficiently.
- **Transparency** — Clear records of booking, procurement and payments.
- **Scalability** — Can expand from one district to multiple districts and states.

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RESEARCH AND REFERENCES

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✓ **MP e-Uparjan** — Government of Madhya Pradesh **https://mpeuparjan.nic.in/**

- **Studied :** Farmer registration, procurement-centre allocation, slot booking.
- **Identified Gaps :** Separate Payment Portal, Limited Slot Flow, Fragmented Journey

✓ **UP e-Mandi** — Uttar Pradesh State Agricultural Produce Markets board **https://emandi.up.gov.in**

- **Studied :** e-Procurement & Mandi Workflow
- **Identified Gaps :** Limited Live-Quote Visibility, Complex UI, Fragmented Journey.

✓ **e-Platform for National Agricultural Market** — NITI Aayog **https://www.niti.gov.in/**

- **Studied :** fragmentation and inefficiencies in agricultural markets and the role of integrated electronic market platforms.

NOVELTY IN IMPLEMENTATION

- **Unified Farmer Journey** → All essential services on a single platform.
- **Smart Slot & Token Management** → Date, time slot and token in one seamless flow.
- **Live Queue Tracking** → Farmers can track their real-time queue position.
- **Payment Transparency** → Payment status available in one place.
- **Multi-State Scalability** → Supports diverse state-wise procurement workflows.